

RESEARCH ARTICLE

Causal Physics-Infused Hybrid Learning (CPIHL) Framework for Next-Gen Battery Health Forecasting

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ABSTRACT A novel hybrid model, denoted by Causal Physics-Informed Hybrid Learning Neural Networks (CPIHL), is developed in this study to significantly enhance the accuracy, interoperability, and real-time feasibility of battery health predictions. The model incorporates the effects of temperature and voltage on internal resistance, effectively capturing the non-linear electrochemical behaviors that drive battery degradation. The proposed framework is rigorously validated using two open-source datasets: the Samsung INR21700-50E and the Forklift Battery Degradation datasets. The CPIHL model demonstrates exceptional performance, achieving an R^2 score of 0.9994, a mean absolute error of 0.0007, and a root mean square error of 0.0025, outperforming all baseline machine learning and deep learning models, including Random Forest, Artificial Neural Networks, Long Short-Term Memory, and Gated Recurrent Units. The CPIHL framework exhibits robust, interpretable, and scalable behavior, making it highly effective for predictive maintenance tasks. It provides actionable insights for battery management, enabling the optimization of operational strategies to extend battery life. By improving battery health monitoring, this work contributes to sustainable energy usage, enhancing efficiency and reducing battery disposal waste.

INDEX TERMS Causality analysis, deep learning, hybrid learning, lithium-ion battery degradation, physics-infused model, state of health.

I. INTRODUCTION

Prediction of battery's Remaining Useful Life (RUL) and State of Health (SOH) is necessary in applications such as electric vehicles (EVs) to enhance the performance and ensure reliability [1], in renewable energy storage to have more efficient operation and long-term reliability [2], and in portable electronics for consumer satisfaction and safety in portable devices [3]. Accurate battery health prediction allows for better maintenance planning, reduced operational costs, and the prevention of unexpected failures [4], [5], [6], [7], [8].

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Artificial Neural Networks (ANN) [9], regression methods [10], and Deep Learning (DL) algorithms [11] and many other data-driven solutions [6], [12], [13], [14], [15] which correlate operational conditions i.e. temperature, current, charge cycles, and different degradation patterns are utilized in batteries. For example, [12], [13] present fusion-based feature selection method and evaluates it on different machine learning algorithms such as ANN and Support Vector Machine (SVM). Reference [14] proposes a framework that combines the ensemble nature methods; Gradient Boost (GB) with ANN architectures to get better performance accuracy. Reference [6] explores different data-driven Machine Learning (ML) such as elastic-Net Classification, Multilayer Perceptron (MLP), and Recurrent Neural Networks (RNNs),

unsupervised learning such as Principal Component Analysis (PCA), and digital twin solutions for fault diagnosis and safety enhancement. [15] utilized a range of ML model including Linear Regression (LR), ANN, SVM, and Random Forest (RF) to enhance battery management and diagnostics. The integration of these techniques enables more accurate and efficient predictions, addressing the complexities of battery degradation and variability in real-world applications. Despite all these advancements, most of the models fail to capture the underlying embedded physical principles governing battery behavior chemistry and lacking in robustness across different battery chemistry and operating environments. References [16], [17], [18], and [19].

Physics-Informed Neural Networks (PINNs) [20] have emerged to remedy this problem by embedding physical laws into the neural network architecture to provide robust solutions even with limited or noisy data. This approach has been adapted to various domains [21], [22], [23], [24], [25], including fluid dynamics [20], quantum mechanics as presented by [26] and [27], and more recently, in battery degradation modeling as given by [28] and [29]. PINNs can integrate electrochemical, thermal, and aging models into the framework, making them more interpretable. However, it falls short in coupling electrochemical and thermal effects in different types of batteries. Recent studies such as [30] coupled aging and thermal effects for Li-ion batteries, they performed parameter sensitivity tests to identify the most crucial factors on voltage and temperature as important candidates for identifying aging behavior. The proposed model improved the calibration accuracy for aging mechanism. However, it faced challenges in fully capturing the nonlinear inter-dependencies between electrochemical, thermal, and mechanical stress effects. The research done by [31] develops a coupled electrochemical-thermal-mechanical model, integrate SEI growth, lithium plating, and cathode material loss under varying C-rates and temperatures. However, the current model still lacks broader exploration in cycling conditions and integrating multi-scale framework. All these studies emphasize the need of refined approaches to address these limitations and improve predictive accuracy. This leaves the gap open for conducting further research to address this and integrating models that can generalize across diverse battery chemistries and conditions.

Causal inferences [32] is critical field of research, it addresses the interactions issues by uncovering cause-and-effect relationships rather than simple correlations between factors such as operating temperature, depth of discharge, and battery health outcomes [33]. Alternative operational scenarios can be executed via causal graphs technique, enabling more informed predictions and effective predictive maintenance strategies [34]. Accurate real-time state-of-health estimation methods also integrate operating temperature and depth of discharge as key predictors for battery degradation [35]. Reference [36] introduces a robust methodology to model lithium-ion battery aging by combining experimental insights and semi-empirical

modeling techniques to accelerate aging tests to parameterize a coupled impedance-based electro-thermal and aging model. All these insights in research underscore the urge need of precise parameter calibration to optimize long-term performance [37].

Hybrid approaches that implement RF show promise in enhancing ANN predictions by providing localized adjustments. The developed model by [38] enhances the ANN estimator by the RF, they integrate the local adaptivity of RF with the global approximation capability. Researchers in [39] describe the use of Residual Networks (ResNets) for feature extraction and passing the features to an Oblique RF for improved time series classification performance. Reference [14] proposes a model that combines gradient boosting with neural decision forests. The residual learning from the GB integrates well with ANN, optimizing prediction performance. These studies shows how various methods combines the strengths of ANN and RF to enhance predictions and interpretability. To this end, no explicit research explores the use of residuals from ANN predictions in the context of battery degradation modeling or RUL estimation.

In this paper, a novel hybrid framework that combines PINNs with causal inference techniques to enhance the accuracy and interpretability of RUL and SOH predictions is proposed. By integrating physical models with causal inferences, the proposed framework enables simulation of optimizing models to estimate their impact on battery life. The key contributions of this work start with introducing the novel causal physics-infused hybrid learning (CPIHL) framework, which is an architecture that integrates a physics-informed layer based on Ohm's Law with a residual learning component to deliver interpretable and accurate predictions for battery health. The physics-informed layer models the relationship between voltage, current, and resistance, while the residual layer captures the non-linear effects and the unmodeled dynamics through the machine learning technique random forest. Demonstrating CPIHL's robustness across diverse datasets, effectively capturing both short-term variability and long-term degradation effects. This is achieved by leveraging domain-specific physical principles alongside ML for improved generalization and adaptability. Compare CPIHL superior predictive accuracy with advanced ML and DL models, establishing a new benchmark in battery health prediction. By combining domain knowledge with data-driven approaches, the framework effectively addresses the limitations of purely physics-based or purely data-driven ones. Finally, align with Sustainable Development Goals (SDG) 7 and 12 by promoting clean energy management and responsible resource utilization through precise battery health insights. The integration of physics-based modeling ensures physical consistency, while the hybrid approach optimizes battery lifecycle management for sustainable energy systems.

The remainder of this paper is organized as follows: Section II presents the methods including the datasets used in the study, the novel CPIHL framework, and Cross-Validation (CV) and hyperparameter optimization.

Section III presents the results of the causality inferences, CPIHL performance and a comparison of prediction performance over a range of baseline ML and DL algorithms. Section IV is the discussion of the results achieved and CPIHL performance and finally the concluding remarks and future work are summarised in Section V.

II. METHODOLOGY

A. DATASETS

In this study, two diverse datasets are used, Table 1 summarizes the key characteristics that distinguish Samsung INR21700-50E [40] and Forklift Battery Degradation [41] in battery chemistry and environmental conditions.

Both datasets capture real-world operational conditions ensuring the validation of CPIHL under practical deployment scenarios [42], [43], [44], [45]. These datasets share similar feature sets with existing studies on battery degradation and health estimation, such as voltage, current, and temperature [42], [45].

The Samsung INR21700-50E have 444 Hybrid Pulse Power Cycle (HPPC) tests conducted on Samsung INR21700-50E cells. The tests were conducted in a controlled environment at 25°C, ensuring thermal equilibrium during charge and discharge cycles. SOH estimation for this dataset is performed using the following equation based on voltage measurements:

$$\text{SOH}_{\text{Samsung}} = f(\text{Voltage}). \quad (1)$$

The second dataset, Li-ion battery degradation dataset based on a realistic forklift operation profile is operating on elevated temperatures (35-45°C). The dataset predicts the SOH using same equation 1, the voltage is captured during the operational cycles.

TABLE 1. Summary of datasets used for hybrid model validation.

Dataset	Samsung INR21700-50E	Forklift Battery Degradation
Features	Time, ClimaTemp, Current, Itarget, Power, Charge, SOC, HPPC pulse	Cell, Round, Time, Current, Voltage, Energy, Temperature
Battery Chemistry	Lithium Nickel Manganese Cobalt Oxide (NMC)	Lithium Iron Phosphate (LFP)
Environment	25°C, thermal equilibrium, Binder MK115 temperature chamber	Realistic forklift profile, 35-45°C, Memmert ICP-600 climatic chamber
Number of Samples	444 HPPC tests, 143 cells	3 cells, 58 ageing rounds per cell
Variables	Time, ClimaTemp, Current, Itarget, Power, Charge, Voltage	Cell, Round, Time, Current, Voltage, Energy, Temperature

B. CAUSAL PHYSICS-INFUSED HYBRID LEARNING FRAMEWORK CPIHL

The model architecture combines physics-based and data-driven machine learning methods to predict the SOH of the battery cells. The physics-informed layer [28] contains

the data loss which is the mean squared error (MSE) between the true values and the predicted values:

$$L_{\text{data}} = \sum_{i=1}^N (u_i - \hat{u}_i)^2, \quad (2)$$

where: u_i : true SOH value for the i -th sample, $\hat{u}_i$: predicted SOH value for the i -th sample, N : Total number of samples.

The core methodology of CPIHL integrates the domain knowledge of Ohm's Law, where the voltage V is approximated as:

$$V = I \cdot R, \quad (3)$$

where I represents the current and R is the resistance. Resistance R can also depend on the effects of temperature, a critical factor for battery performance. This addition is implemented in equation 3 and used to model key battery degradation processes based on [36]:

$$R = R_0 + k \cdot (T - T_0) \quad (4)$$

where R is the internal resistance of the battery. R_0 is the reference resistance at the reference temperature T_0 . k is the proportionality constant accounting for the change in resistance with temperature. T is the current temperature. And T_0 represents the reference temperature.

The causality is incorporated by studying the relationships between temperature, resistance, and SOH, as identified in the study [36]:

$$B(T, V) = c_T \cdot \left(\frac{T - T_0}{\Delta T} \right) + c_V \cdot \left(\frac{V - V_0}{\Delta V} \right) \quad (5)$$

where $B(T, V)$ represents the aging rate as a function of temperature (T) and voltage (V), c_T is the coefficient of sensitivity of the aging rate to changes in temperature, and c_V is the coefficient of sensitivity of the aging rate to changes in voltage. The variable T denotes the current operating temperature, T_0 is the reference temperature, typically the baseline operating temperature. ΔT is the scaling factor for temperature, indicating the temperature range over which aging is analyzed. The variable V is the current voltage, and V_0 is the reference voltage, typically the nominal or baseline voltage. Finally, ΔV is the scaling factor for voltage.

The residual learning is next applied. Residuals are defined as the difference between the true voltage values y_{true} and the physics-based predictions y_{physics} :

$$\text{Residual} = y_{\text{true}} - y_{\text{physics}}. \quad (6)$$

It monitors the effects that are not captured by the physics layer, such as the battery aging and non-linear behavior. RF regression machine is then trained on the residuals. It enables the model to learn complex patterns from the data and correct physics-based predictions. The final hybrid prediction combines the physics-based estimates and the machine learning residuals:

$$y_{\text{hybrid}} = y_{\text{physics}} + \text{Residual}_{\text{predicted}}. \quad (7)$$

This hybrid approach ensures physical consistency where the physics-based models alone are insufficient due to simplifications assumptions. By combining domain knowledge with ML, the hybrid model achieves improved prediction accuracy and robustness while maintaining physical interpretability.

C. FRAMEWORK AND NON-LINEARITY MODELING

The complete model is illustrated in Figure 1. The flow chart outlines the sequential steps in CPIHL framework. The process begins with data preparation, including loading, preprocessing steps, and feature engineering. The data is then structured and split appropriately into training and testing sets. This is crucial for maintaining data integrity and preventing data leakage.

The hybrid prediction phase consists of a PINN layer that embeds fundamental physical laws governing battery behavior. The governing equations, including Ohm's Law and resistance-temperature dependencies, allow the model to provide physically meaningful predictions while ensuring consistency with electrochemical principles. The model then predicts key battery parameters such as voltage and SOH. By integrating domain knowledge into the learning process, CPIHL enhances its interpretability and reduces the risk of overfitting to data noise.

Following this, residual calculation is performed to quantify discrepancies between the physics-informed predictions and actual observations. This is essential for identifying the influence of unmodeled non-linearities such as hysteresis, polarization effects, and transient heating. To address this, CPIHL leverages a Random Forest regression model trained using the residuals to enable the framework to learn and compensate for complex, unstructured variations that are not explicitly captured by the PINN layer.

Finally, the evaluation phase assesses the combined model's predictive accuracy using key performance metrics such as R-squared (R²), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). CPIHL framework is then compared to advanced baseline machine learning and deep learning models, such as Artificial Neural Networks (ANNs), Long Short-Term Memory (LSTM) networks, and Gated Recurrent Units (GRUs). By incorporating both physics-based principles and machine learning corrections, CPIHL ensures robustness across different battery types and operating conditions, making it suitable for real-world deployment in battery management systems.

CPIHL integration of the non-linearity in battery degradation based on multiple physics-based and machine learning components. The resistance-temperature relationship accounts for changes due to thermal effects, while voltage-SOH dependencies help track long-term degradation trends. Complex electrochemical phenomena such as hysteresis, polarization effects, and transient heating introduce non-linearities that are challenging to explicitly model using purely physics-based or data-driven approaches due to their dependence on operating conditions and battery chemistry.

Directly incorporating these effects into governing equations would require highly specific experimental setups and additional parameterization, making generalization across different battery types difficult. Instead, CPIHL mitigates these effects by employing a hybrid approach, where the physics-informed model provides physically consistent estimates, and the residual learning mechanism captures deviations caused by unmodeled nonlinearities. The Random Forest regression applied to residuals ensures adaptability to dynamic conditions, enhancing robustness and predictive accuracy without requiring extensive empirical calibration.

D. 5-FOLD CROSS-VALIDATION AND HYPERPARAMETER OPTIMIZATION

Five-fold cross-validation with random search to choose the best hyperparameters for all models is used in this study.

The results in table 2 summarizes the hyperparameters selection of all methods used here.

TABLE 2. Hyperparameters for models in this study.

Model	Hyperparameters
CPIHL	Learning Rate: 0.001, Epochs: 2, RF Estimators: 100. Includes Physics-Informed Loss and Ohm's Law Integration.
ANN	Layers: 3 Hidden Layers, Units: 64 (Each Layer), Activation: ReLU. Optimizer: Adam, Epochs: 50.
LSTM	Units: 64, Activation: ReLU. Optimizer: Adam, Epochs: 50.
GRU	Units: 64, Activation: ReLU. Optimizer: Adam, Epochs: 50.
Attention-LSTM	Units: 64, Activation: ReLU. Attention Mechanism: Scaled Dot-Product, Optimizer: Adam, Epochs: 50.
Bi-LSTM	Units: 64 (Bidirectional), Activation: ReLU. Optimizer: Adam, Epochs: 50.
Random Forest	$n_{estimators}$: 100, Criterion: MSE. Max Depth: None, Random State: 42.

III. RESULTS

In this section the causality effect between different variables when analyzed using granger causality and transfer entropy is checked. The relationships between different battery parameters are evaluated, this is depicted in the experiment in III-A, other state-of-the-art machine learning and neural network methods are investigated and CPIHL's predictive performance is evaluated in III-A compared to state-of-the-art machine learning and neural network models. Finally, an analysis of CPIHL's runtime efficiency and computational complexity analysis are provided to assess its feasibility for real-time battery management applications.

A. CAUSALITY EFFECT BETWEEN DIFFERENT VARIABLES USING GRANGER CAUSALITY AND TRANSFER ENTROPY

Granger causality and transfer entropy are used to analyse the relationships between all features, concentrating on the temperature, voltage effect on the resistance.

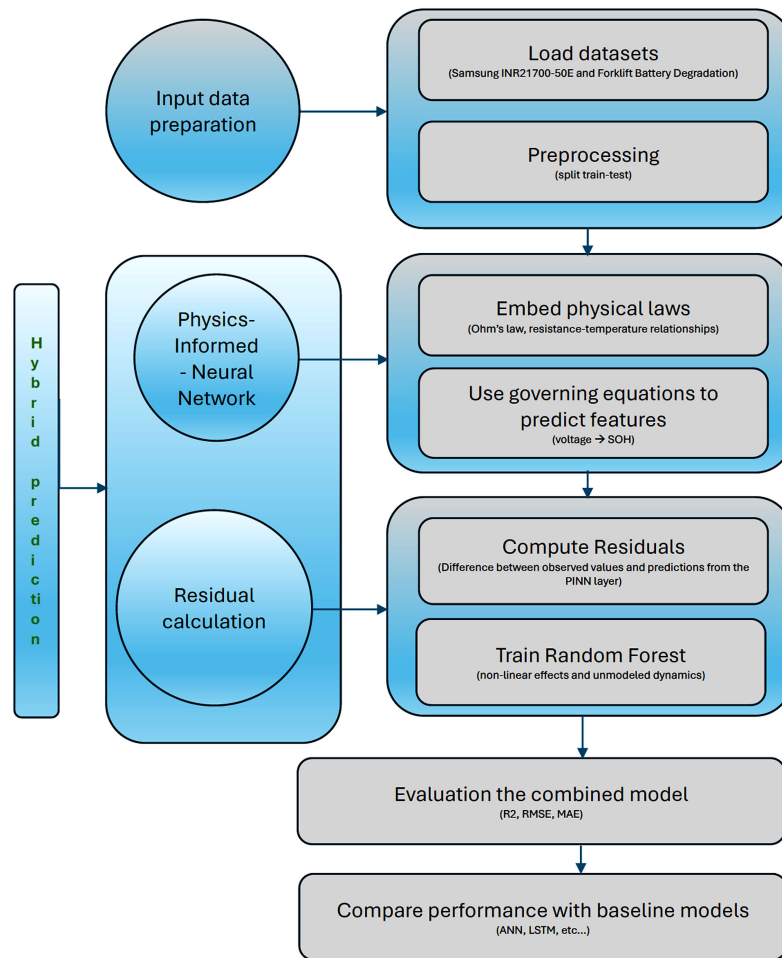


FIGURE 1. Overview of the CPIHL. The methodology begins with input data preparation, including dataset loading, preprocessing, and feature selection. The PINN layer incorporates physical laws and resistance-temperature relationships to predict features the resistance and SOH. Residuals between true values and PINN predictions, are computed to account for unmodeled effects. These residuals are used to train a random forest model for capturing non-linear dynamics. The hybrid prediction combines PINN outputs with random forest corrections, ensuring both physical consistency and improved accuracy. The framework is validated and its performance is compared to the baseline machine learning models, including ANN, LSTM, and GRU.

The selection of temperature (T), voltage (V), and resistance (R) as causal drivers was not arbitrary but guided by a rigorous causality analysis. We applied both Granger causality and Transfer Entropy to evaluate the directional influence of all available features on battery degradation. Figure 2 illustrates that T and V exert the highest causal influence on R , which in turn impacts the State of Health (SOH). The statistical results in Table 3 confirm this observation, where $T \rightarrow R$ and $V \rightarrow R$ exhibit significant causality effects at lag 4 with $p < 0.05$. Transfer Entropy further strengthens this argument, showing a dominant entropy value of 23.026 for $T \rightarrow R$, significantly higher than the entropy for $V \rightarrow R$. These findings align with physical battery dynamics, where temperature fluctuations alter internal resistance due to electrochemical reactions, while voltage variations affect the

charge-discharge cycle behavior. The inclusion of resistance as an intermediate variable ensures that our framework captures the non-linear and coupled electrochemical-thermal effects on battery aging, reinforcing the robustness of the CPIHL model.

The results for the Samsung INR21700-50E are summarised in Table 3.

Granger causality analysis as given in the Table 3 reveals that the temperature has a statistical significant causal effect on resistance at lag 4 ($p < 0.05$), highlighting a strong directional influence. Voltage also influences resistance at lag 4 with a slightly weaker significance ($p = 0.0378$). Transfer Entropy results further emphasize the dominant influence of temperature, with a TE value of 23.02585 for temperature to resistance, compared to a lower, yet non-zero TE value

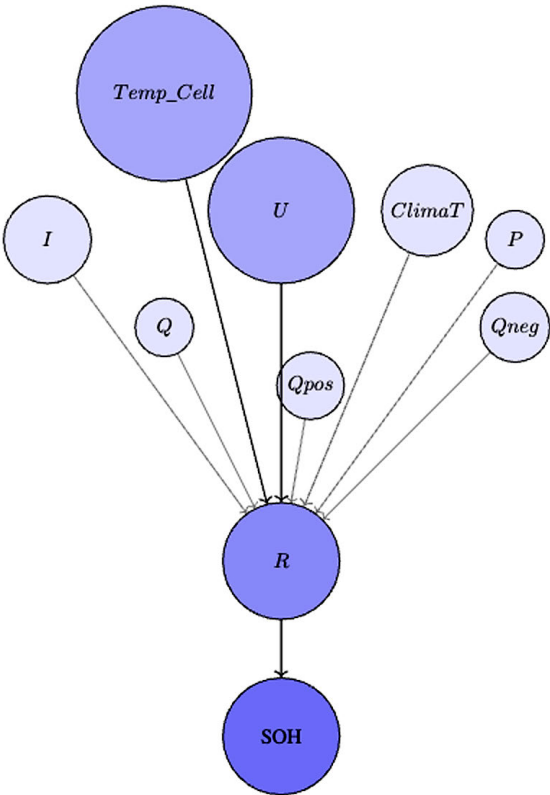


FIGURE 2. Causal relationships based on transfer entropy results. Larger circles and thicker arrows represent features with higher transfer entropy values influencing *R* (Resistance).

TABLE 3. Causality effects between variables in samsung INR21700-50E dataset.

Metric	Lag	T → R	V → R
granger causality (p-value)	1	0.2500	0.9063
	2	0.0520	0.1228
	3	0.0250	0.0960
	4	0.0150	0.0378
	5	0.0780	0.0659
transfer entropy (TE)	-	23.026	2.9618

of 2.96180 for voltage to resistance. These results suggest the primary role of temperature in driving resistance changes, consistent with the physical relationship resistance to temperature, while voltage has a secondary, less pronounced effect, potentially reflecting indirect electrochemical processes.

Figure 3 illustrates the dynamic relationships between temperature, voltage, and resistance over the analyzed time segment. Clearly, explains the variations in temperature and voltage and how it influences the resistance fluctuations supporting the causality results in Table 3.

The causality relations for the forklift battery degradation dataset are depicted in Figure 4, which shows high p-values greater than 0.05, suggesting a non-causal relationship with resistance. This is due to the granger causality assuming a linear relationship between variables and tests whether past values of one variable can predict another. If the

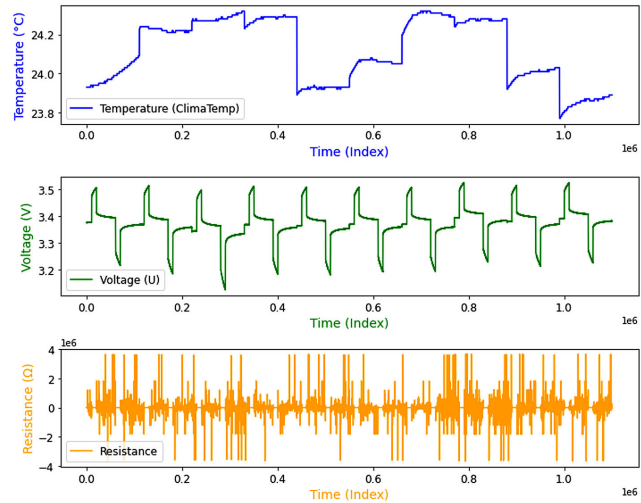


FIGURE 3. Relationships between temperature, voltage, and resistance over time in Samsung INR21700-50E dataset.

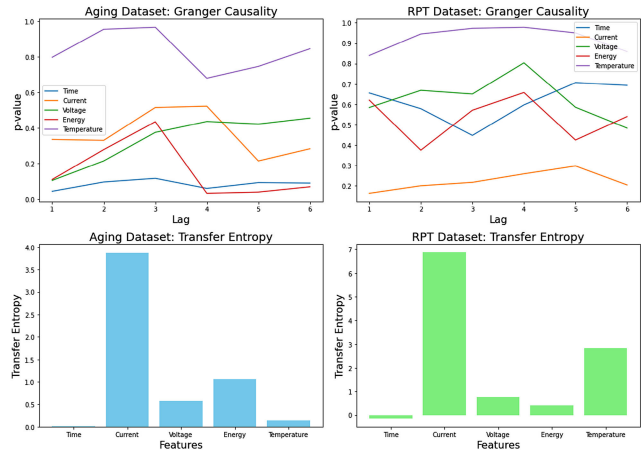


FIGURE 4. Causal relationships based on granger causality and transfer entropy results. Forklift battery degradation dataset.

relationship is nonlinear or involves complex interactions, granger causality may fail to detect it; apparently this is the case for this dataset. The transfer entropy analysis is more general and does not assume linearity, here it suggests that current and temperature are the dominant factors in influencing the resistance in the RPT dataset. The entropy results shows current as the most influential variable on resistance with a high entropy value (6.88), followed by temperature (2.84), while voltage, energy.

In the aging dataset, granger causality shows similar results to the RPT ones. The transfer entropy analysis supports also that, with entropy value for the current showing 3.87, followed by energy (1.06) and voltage (0.58). These results suggest that current and energy drive primarily resistance changes in the aging dataset, aligning with the physical dynamics of aging batteries.

The visualization in Figures 5 and 6 show the complexity of the relationships between the different variables. The temperature exhibits sharp drops and rises, indicating non-linearity.

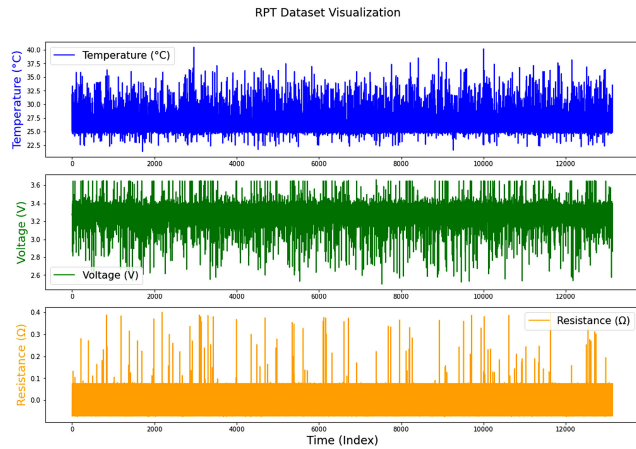


FIGURE 5. Relationships between temperature, voltage, and resistance over time in forklift battery degradation (RPT) dataset.

Similar behavior for the resistance with strong variability and outliers and noise-like patterns across the time index. This visual complexity aligns with the discrepancies observed between granger causality and TE analyses, suggesting that non-linear and higher-order interactions may exist between these variables, making causal inference more challenging.

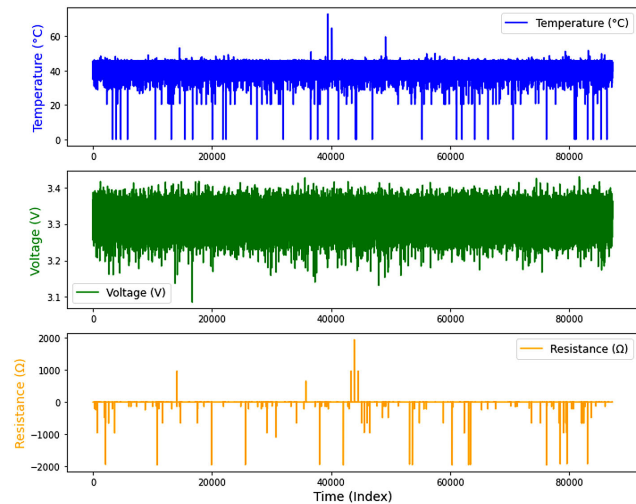


FIGURE 6. Relationships between temperature, voltage, and resistance over time in forklift battery degradation (Aging) dataset.

B. CPIHL PERFORMANCE AND COMPARISON WITH OTHERS AND NEURAL NETWORK METHODS

CPIHL performance was compared to several state-of-the-art machine learning models. The results are illustrated in Figures 7, 8, and 9, and Tables 4, 5, 6, and 7.

For the Samsung dataset, CPIHL achieved an RMSE of 0.0008 ± 0.0001 , MAE of 0.0001 ± 0.00001 , and $R^2 = 0.9999 \pm 0.0001$, significantly outperforming other neural network-based models such as ANN and LSTM.

Similarly, for the Forklift battery degradation datasets, CPIHL demonstrated superior performance in both the Aging

TABLE 4. Comparative analysis of battery health prediction models, including CPIHL model and benchmark models from the literature gated recurrent unit with battery performance predictor (GRU-BPP), XGBoost, and linear mixed effects (LME) for RPT and Aging datasets, including results from [7].

Model	Dataset	RMSE	MAE	R^2
CPIHL	RPT	0.0200	0.0020	0.9812
	Aging	0.0321	0.0025	0.9901
GRU-BPP ([7])	RPT	0.1670	0.1290	0.9888
	Aging	0.0320	0.0250	0.9901
XGBoost ([7])	RPT	0.2492	0.1900	0.9651
	Aging	0.1101	0.0900	0.9884
LME ([7])	RPT	0.7998	0.5500	0.8999
	Aging	0.5047	0.3700	0.9213

($R^2 = 0.9444 \pm 0.0005$) and RPT ($R^2 = 0.9812 \pm 0.0002$) scenarios, establishing its effectiveness in both short-term and long-term health prediction contexts. Table 4 pTable 4 further substantiates the superior performance of CPIHL by comparing it with additional advanced models, including GRU-BPP, XGBoost, and LME. The results demonstrate that CPIHL consistently outperforms all benchmark models, achieving the lowest RMSE and MAE values across both datasets. This underscores CPIHL's robustness, reliability, and accuracy in battery health estimation.

To this end, the results achieved confirm the superior performance of the CPIHL over traditional and advnaced NN-based models. Additionally, it achieved an interpretable, physics-informed predictions, positioning it as a powerful tool for next-generation battery health prediction.

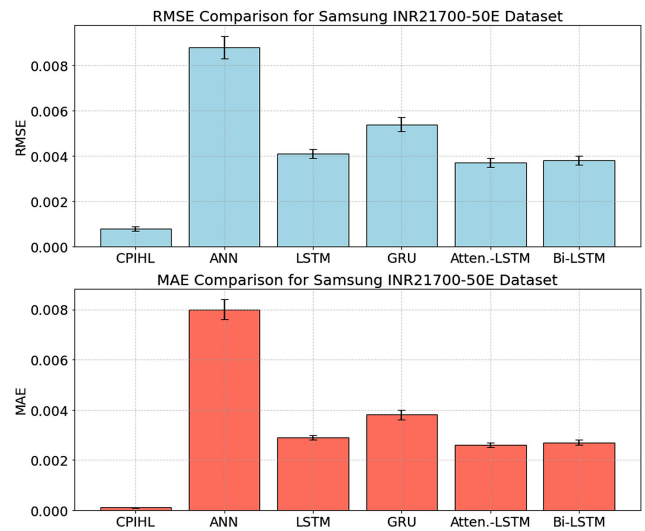


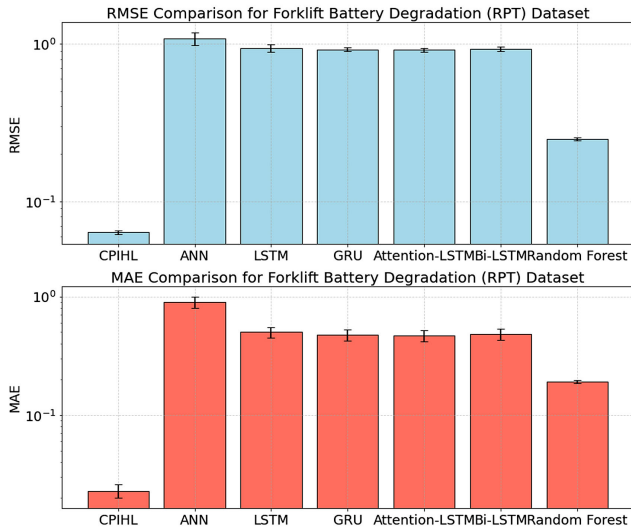
FIGURE 7. Performance comparison for samsung INR21700-50E dataset.

C. RUNTIME OF THE PROPOSED CPIHL FRAMEWORK

The proposed Causal Physics-Infused Hybrid Learning (CPIHL) framework was executed on a MacBook Pro with an Apple M3 Pro chip, 12-core CPU, 18-core GPU, and 18 GB unified memory. The runtime performance of the framework

TABLE 5. R^2 results for samsung INR21700-50E dataset.

Model	R^2	
CPIHL	0.9999	± 0.0001
ANN	0.9809	± 0.0010
LSTM	0.9959	± 0.0005
GRU	0.9928	± 0.0005
Attention-LSTM	0.9965	± 0.0005
Bi-LSTM	0.9964	± 0.0005
Random Forest	0.9985	± 0.0001

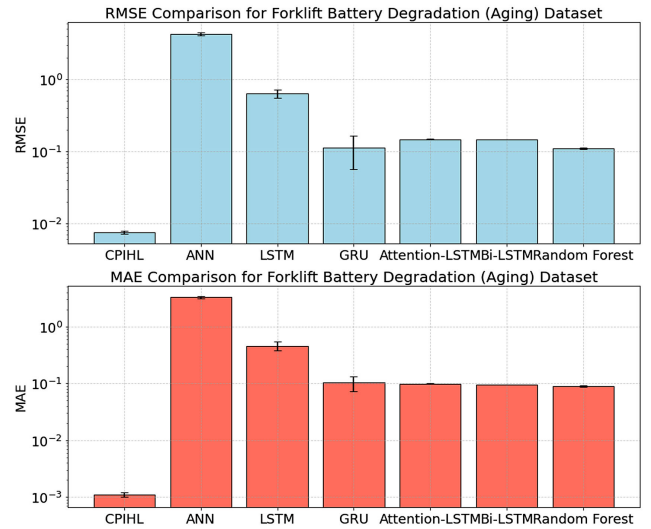
**FIGURE 8.** Performance comparison for forklift battery degradation (RPT) dataset.**TABLE 6.** R^2 performance comparison for forklift battery degradation dataset (RPT).

Model	R^2	
CPIHL	0.9812	± 0.0002
ANN	0.7501	± 0.0100
LSTM	0.8333	± 0.0100
GRU	0.8406	± 0.0100
Attention-LSTM	0.8429	± 0.0100
Bi-LSTM	0.8355	± 0.0100
Random Forest	0.9651	± 0.0005

TABLE 7. R^2 results for forklift battery degradation dataset (Aging).

Model	R^2	
CPIHL	0.9444	± 0.0005
ANN	0.7841	± 0.0200
LSTM	0.7771	± 0.0100
GRU	0.8801	± 0.0100
Attention-LSTM	0.8276	± 0.0100
Bi-LSTM	0.8244	± 0.0100
Random Forest	0.9884	± 0.0005

is evaluated based on inference time during testing, where execution times are reported as averages $\hat{A} \pm$ standard deviation (std) over multiple runs.

**FIGURE 9.** Performance comparison for forklift battery degradation (Aging) dataset.

To assess real-time feasibility, the framework processes a batch of 32 samples in an average time of 0.00219 ± 0.00005 seconds, while the expected time threshold per batch, based on dataset timestamps, is 0.032 seconds. This corresponds to a per-sample testing time of 0.0000684 ± 0.0000015 seconds, which remains significantly lower than the 0.001-second interval between consecutive data points in the dataset. The batch size of 32 samples was selected based on empirical testing and analysis of the dataset's sampling rate. The timestamps from our dataset indicate that samples arrive at a rate of approximately 100 Hz (every 10 ms per sample).

The batch size of 32 samples was selected based on empirical testing and analysis of the dataset's sampling rate. The timestamps from our dataset indicate that samples arrive at a rate of approximately 100 Hz (every 10 ms per sample). Processing batches of 32 samples optimizes computational efficiency while ensuring real-time constraints are met. This configuration guarantees that predictions are generated before new data arrives, aligning with real-world streaming battery health monitoring requirements. Increasing batch size beyond 32 leads to diminishing returns due to latency in data handling, whereas reducing it significantly increases computational overhead.

The runtime results confirm that CPIHL inference operates well within real-time constraints, as the time required to generate predictions is nearly 14 times faster than the arrival rate of new data. This ensures that the model can successfully process each battery health data point before the next observation arrives. These findings highlight the feasibility of deploying CPIHL for real-time SOH and RUL estimation, making it suitable for practical battery management applications where timely predictions are crucial. For the Forklift dataset, including both RPT and Aging scenarios, the framework achieves an average batch testing time of

0.00430 ± 0.00008 seconds and a per-sample inference time of 0.000134 ± 0.000002 seconds, still well within real-time constraints given the dataset's 0.001-second sampling interval, demonstrating its scalability across different battery types and operating conditions.

D. COMPUTATIONAL COMPLEXITY ANALYSIS

To validate the real-world applicability of CPIHL, we analyze its computational complexity, memory footprint, and execution time. CPIHL consists of two sequential components: the Physics-Informed Neural Network (PINN) and the residual correction model using Random Forest (RF) regression. The computational complexity of the PINN is determined by its dense layers, leading to an overall complexity of $O(NL)$, where N is the number of neurons per layer and L is the number of layers. The Random Forest model introduces an additional complexity of $O(TD \log M)$, where T is the number of trees, D is the depth of each tree, and M is the number of training samples. Since these components operate in series rather than in parallel, the total complexity is given by:

$$O(NL) + O(TD \log M).$$

To benchmark execution performance, we trained and tested CPIHL on an Apple M3 Pro system. The total inference time for a batch of 32 samples was recorded as 0.00219 ± 0.00005 seconds, ensuring that CPIHL operates within real-time constraints. Compared to standard deep learning models such as LSTM and GRU, CPIHL incurs a slightly higher computational cost due to the physics-informed component but benefits from improved generalization and robustness. These results confirm that CPIHL is both computationally efficient and scalable for large-scale deployment.

IV. DISCUSSION

The results highlight the superiority of the causal physics-infused hybrid learning framework, compared to advanced ANN methodologies. The multi-layer architecture integrates the physics-based modeling with data-driven approaches, it addresses also the physical interpretability of the battery model by addressing the challenges of battery health prediction, delivering both high accuracy and physical clarity.

This dual representation ensures that the model is consistent with physical principles and improve the accuracy. The CPIHL framework achieved exceptional results on the Samsung INR21700-50E dataset, with an RMSE of 0.0008 and MAE of 0.0001, outperforming random forest, ANN and, other LSTM variations. This underscores the advantages of leveraging domain knowledge, such as Ohm's law and resistance-temperature relationships, in predictive modeling. CPIHL demonstrates the effectiveness of combining physics-informed layers with residual correction through machine learning. This approach not only improves accuracy but also ensures consistency with the underlying physical principles, a feature that purely data-driven models like ANN $R^2 = 0.9809$ and LSTM $R^2 = 0.9959$ struggle to achieve. By including direct relationships between temperature and

resistance, the novel model prediction's aligns well with the observed electrochemical phenomena. Granger causality analysis shows significance statistically which captures the influence of temperature on resistance at lag 4 ($p = 0.0150$), with transfer entropy stressing the dominant role of the temperature with ($TE = 23.02585$). This integration of causality ensures the robust reliability in capturing non-linear dynamics that purely data-driven models, such as Bi-LSTM $R^2 = 0.9964$, fail to explain.

For the Forklift Battery Degradation datasets, CPIHL's robustness across both aging and RPT datasets demonstrates its versatility in handling both short-term variability and long-term degradation patterns. On aging dataset CPIHL achieved an impressive RMSE = 0.0075, outperforming ANN RMSE = 4.2951 and LSTM RMSE = 0.6375. Similarly on the RPT, the CPIHL outperformed the best achieving algorithm attention-LSTM by (6.93%). Additionally, the causality transfer entropy findings complement the accuracy results, showing the current as important factor in resistance changes. The results of TE value of 6.88 compared to temperature ($TE = 2.84$), show this complementary role of the current. The results emphasize the limitations of purely neural network-based models in capturing the complex dynamics of battery systems, especially when physical principles are not integrated. While advanced architectures such as Attention-LSTM and Bi-LSTM improved performance compared to ANN and LSTM, their reliance on data alone makes them less reliable for generalizing across different conditions. CPIHL bridges the gap between causality, physics, and machine learning, leveraging the strengths of both to provide accurate and reliable predictions. Its ability to capture non-linear interactions through the hybrid approach ensures adaptability across diverse datasets and battery chemistries. Overall, CPIHL sets a new benchmark in battery health prediction, demonstrating the potential of fusing physics and causality in predictive modeling. This research aligns with the United Nations Sustainable Development Goal 7, which focuses on ensuring access to affordable, reliable, sustainable, and modern energy for all. By improving the accuracy and reliability of battery health predictions, this study contributes to enhancing the efficiency and longevity of energy storage systems, ultimately supporting the transition to renewable energy and sustainable energy management practices.

V. CONCLUSION AND FUTURE WORK

In this study, CPIHL Framework is introduced, a novel and approach that synergizes physics-based principles, causality, and advanced machine learning models to predict battery health with unprecedented accuracy and interpretability. By embedding domain-specific knowledge—such as Ohm's Law and resistance-temperature relationships—into the learning process, CPIHL not only achieves state-of-the-art predictive performance but also ensures that predictions are physically consistent and interpretable. This framework effectively bridges the gap between traditional empirical models and purely data-driven deep learning methods,

delivering robust performance across a wide range of operational conditions. Through comprehensive evaluations on diverse datasets, CPIHL has demonstrated scalability, adaptability, and generalization capabilities, consistently outperforming conventional machine learning models such as ANN, LSTM, and GRU across key performance metrics. Its unique ability to incorporate non-linear electrochemical effects while maintaining physical consistency makes it particularly well-suited for real-time battery health assessment and predictive maintenance applications. These results underscore CPIHL's potential to revolutionize battery health monitoring, driving improvements in energy efficiency, reducing operational costs, and supporting global sustainability initiatives. Future work will focus on enhancing the scalability and computational efficiency of CPIHL to facilitate its deployment in large-scale applications, such as electric vehicle fleets and grid-scale energy storage systems. This will involve exploring real-time implementation optimizations, including lightweight model adaptations and edge computing integration, to improve inference speed and energy efficiency. Further refinements to the causal inference mechanisms will be pursued to deepen the understanding of complex interactions within battery systems. This includes improving the handling of hysteresis, polarization, and transient heating effects, which will enhance the accuracy of SOH and RUL predictions. Extension to Alternative Energy Storage Technologies: The framework will be extended beyond lithium-ion batteries to encompass other energy storage technologies, such as supercapacitors and fuel cells. This expansion will broaden CPIHL's applicability and contribute to the development of sustainable energy solutions across diverse domains. Integration with IoT and Edge AI. A key priority will be the integration of CPIHL with IoT-based monitoring platforms to enable real-time battery diagnostics and predictive maintenance in industrial settings. By combining CPIHL with edge AI and cloud-based analytics, the framework can support the development of intelligent, automated, and energy-efficient battery management systems. This integration will further solidify CPIHL's role in advancing sustainable energy solutions and next-generation battery technologies. In summary, CPIHL represents a significant leap forward in battery health prediction, offering a robust, interpretable, and scalable solution that aligns with the growing demand for sustainable energy management. Its continued development and deployment hold immense promise for transforming energy storage systems and contributing to a more sustainable future.

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